

Experiential Learning Exercises

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Question 1: Bank Transaction – CRUD Operations (MongoDB)

A bank maintains a transactions collection with the following structure:

```
{  
  "transactionId": "TXN1001",  
  "accountNumber": "ACC12345",  
  "customerName": "Ravi Kumar",  
  "transactionType": "Credit",  
  "amount": 15000,  
  "transactionDate": "2024-02-15",  
  "balance": 45000  
}
```

(a) Write MongoDB queries for the following operations:

- 1. Insert a new transaction record for a customer.**
- 2. Retrieve all transactions of a given account number.**
- 3. Update the balance after a debit transaction.**
- 4. Delete a transaction using transactionId.**

(b) Write the equivalent SQL queries for the above operations assuming a table named Transactions.

Question 2: Aggregation in MongoDB

A bank stores all transaction records in a collection named transactions with the following fields:{

```
"transactionId": "TXN1001",  
"accountNumber": "ACC101",  
"customerName": "Ravi Kumar",  
"transactionType": "Credit",  
"amount": 5000,  
"branch": "Bangalore",  
"transactionDate": "2024-03-10"  
}
```

Answer the following using MongoDB Aggregation Framework:

- a. Write a query to find the total transaction amount for each account number.**
- b. Write a query to display all transactions belonging to the Bangalore branch.**
- c. Write a query to count the total number of transactions for each account.**
- d. Write a query to display the top 3 accounts with the highest total transaction amount.**